

Multiple Choice

1. **Answer:** A
Options: A) 0.75; B) 0.85; C) 0.45; D) 0.35; E) 0.55
Note: Correct option: A - 0.75
2. **Answer:** B
Options: A) 560; B) 1140; C) 1000; D) 680; E) 760
Note: Correct option: B - 1140
3. **Answer:** A
Options: A) 44; B) 42.9; C) 44.1; D) 42.8; E) 43.29
Note: Correct option: A - 44
4. **Answer:** A
Options: A) $y = x + 3$; B) $y = x + 2$; C) $y = 2x$; D) $y = x - 1$; E) $y = x - 2$
Note: Correct option: A - $y = x + 3$
5. **Answer:** A
Options: A) 512; B) 1024; C) 128; D) 72; E) 32
Note: Correct option: A - 512

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '20'.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'There are 8 groups of marbles with 7 marbles in each group.'.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '8-2 i'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '4'.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '1.1 over 11'.

Long Form Response

11. **Answer:** Let $z = a + bi$, where a and b are real numbers. Then $z^2 = (a + bi)^2 = a^2 + 2abi - b^2$ and $|z|^2 = a^2 + b^2$, so

$$a^2 + 2abi - b^2 + a^2 + b^2 = 3 - 5i.$$

Equating real and imaginary parts, we get $2a^2 - b^2 = 3$,
 $2ab = -5$. From the first equation, $a^2 = \frac{3}{2}$. From the second equation,

$$b = -\frac{5}{2a},$$

so

$$b^2 = \frac{25}{4a^2} = \frac{25}{4 \cdot 3/2} = \frac{25}{6}.$$

Therefore,

$$|z|^2 = a^2 + b^2 = \frac{3}{2} + \frac{25}{6} = \frac{17}{3}.$$

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 675 feet. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key